ORF 524 — Final Exam

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December 12, 2022

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Notes:

- Duration of examination: 180 minutes.
  An additional 15 minutes are given to upload your solutions to Gradescope.
- All parts are equally weighted (5 points).
  Please provide as much detail as possible in your answers.
  Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
  A single PDF file must be submitted to Gradescope.
- Only one page (double-sided) of personal notes are allowed during the examination.
  This is otherwise a closed-book and individual examination.

Question 1: Semiparametric Bandwidth Selection (50 Points)

This question discusses bandwidth selection for kernel-based semiparametric and nonparametric estimators. Let x_i for $1 \leq i \leq n$ be i.i.d. real-valued random variables with Lebesgue density $f(x)$. Consider estimating the average density $\theta = \mathbb{E}[f(x_i)]$ using

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^{n} \hat{f}_{i,n}(x_i), \qquad \hat{f}_{i,n}(x) = \frac{1}{n-1} \sum_{j=1, j \neq i}^{n} \frac{1}{h} K\left(\frac{x_j - x}{h}\right),$$

where $K : \mathbb{R} \to \mathbb{R}$ is an even, bounded and compactly supported kernel function with $\int_{\mathbb{R}} K(u)\,\mathrm{d}u = 1$, and h is a positive bandwidth sequence with $h \to 0$ as $n \to \infty$. The goal of this question is to derive a data-driven procedure for selecting the bandwidth h. We take a plug-in approach based on mean squared error. Recall that the Hoeffding decomposition gives $\hat{\theta}_n = \theta_n + \bar{L}_n + \bar{Q}_n$ where

$$\theta_n = \mathbb{E}[U_n(x_i, x_j)], \qquad U_n(x_i, x_j) = \frac{1}{h} K\Big(\frac{x_i - x_j}{h}\Big),$$

$$\bar{L}_n = \frac{1}{n} \sum_{i=1}^{n} L_n(x_i), \qquad L_n(x_i) = 2\big(\mathbb{E}[U_n(x_i, x_j) \mid x_i] - \theta_n\big),$$

$$\bar{Q}_n = \binom{n}{2}^{-1} \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} Q_n(x_i, x_j), \qquad Q_n(x_i, x_j) = U_n(x_i, x_j) - \frac{1}{2} L_n(x_i) - \frac{1}{2} L_n(x_j) - \theta_n.$$

1. Show that $\mathbb{E}[\bar{L}_n] = 0$, $\mathbb{E}[\bar{Q}_n] = 0$ and $\mathbb{E}[\bar{L}_n \bar{Q}_n] = 0$.
2. Show that $\mathbb{E}[U_n(x_i, x_j) \mid x_i] = \int_{\mathbb{R}} K(u) f(x_i + uh)\,\mathrm{d}u$ almost surely.
3. Suppose the kernel K is of order $P \geq 2$, that is, $\int_{\mathbb{R}} u^p K(u) \mathrm{d}u = 0$ for all $1 \leq p \leq P - 1$. Provide conditions on f such that $\mathbb{E}\big[\hat{\theta}_n\big] = \theta + h^P \mathcal{B} + o(h^P)$ and identify the bias $\mathcal{B}$.
4. Show that $\mathbb{V}\big[\bar{L}_n\big] = \frac{\sigma^2}{n} + O\left(\frac{h^P}{n}\right)$ and give the value of σ^2.
5. Show that $\mathbb{V}\big[\bar{Q}_n\big] = \binom{n}{2}^{-1} \frac{\mathcal{V}}{h} + o\left(\frac{1}{n^2 h}\right)$ and give the value of $\mathcal{V}$.
6. Show that

$$\mathbb{E}\big[(\hat{\theta}_n - \theta)^2\big] = \frac{\sigma^2}{n} + \binom{n}{2}^{-1} \frac{\mathcal{V}}{h} + h^{2P} \mathcal{B}^2 + O\left(\frac{h^P}{n}\right) + o\left(\frac{1}{n^2 h} + h^{2P}\right).$$

   (i) Provide sufficient conditions on h so that $\sqrt{n}(\hat{\theta}_n - \theta) = O_{\mathbb{P}}(1)$.

   (ii) Provide sufficient conditions on h so that $\sqrt{n^2 h}(\hat{\theta}_n - \theta) = O_{\mathbb{P}}(1)$.

7. Give conditions on h so that $\sqrt{n}(\hat{\theta} - \theta) \rightsquigarrow \mathcal{N}(0, V_\theta)$ and find the asymptotic variance V_θ.
8. Find the mean squared error optimal bandwidth for the average density estimator $\hat{\theta}_n$, denoting it by $h_{\texttt{AD}}$. How might you feasibly estimate this in practice?

9. Now consider the integrated mean squared error of estimating $f(x)$ using $\hat{f}_{i,n}(x)$. Assuming for simplicity that f is compactly supported, show that

$$\int_{\mathbb{R}} \mathbb{E}\left[\left(\hat{f}_{i,n}(x) - f(x)\right)^2\right] \mathrm{d}x = h^{2P} B^2 + \frac{V}{(n-1)h} + o\left(\frac{1}{nh} + h^{2P}\right)$$

and identify the bias B and variance V. Show that the integrated mean squared error optimal bandwidth for the nonparametric first-step, leave-one-out kernel density estimator $\hat{f}_{i,n}(x)$ is

$$h_{\mathtt{KD}} = \left(\frac{V}{2PB^2(n-1)}\right)^{\frac{1}{2P+1}}.$$

How would you implement this in practice?

10. Which of the bandwidth selectors $h_{\mathtt{AD}}$ and $h_{\mathtt{KD}}$ is asymptotically larger? For each of $h_{\mathtt{AD}}$ and $h_{\mathtt{KD}}$, explain whether the asymptotic normality result in part 7 holds.

Question 2: Optimal Conditional Z-estimation (50 Points)

This question discusses optimal Z-estimation and inference using conditional estimating equations (i.e., moment conditions or score restrictions). Let $(\mathbf{z}_i, \mathbf{x}_i)$ for $1 \leq i \leq n$ be i.i.d. taking values in $\mathbb{R}^{d_z} \times \mathbb{R}^{d_x}$. Assume that

$$\mathbb{E}_{\boldsymbol{\theta}_0}[m(\mathbf{z}_i, \boldsymbol{\theta}) \mid \mathbf{x}_i] = 0 \text{ a.s.} \qquad \Longleftrightarrow \qquad \boldsymbol{\theta} = \boldsymbol{\theta}_0,$$

where $\boldsymbol{\theta} \in \boldsymbol{\Theta} \subseteq \mathbb{R}^d$ and $m : \mathbb{R}^{d_z} \times \mathbb{R}^d \to \mathbb{R}$ satisfies $\mathbb{E}\left[m(\mathbf{z}_i, \boldsymbol{\theta})^2\right] < \infty$ for all $\boldsymbol{\theta} \in \boldsymbol{\Theta}$.

1. Let $\mathcal{G}$ be the class of measurable functions $\mathbf{g}$ from $\mathbb{R}^{d_x}$ to $\mathbb{R}^d$ which satisfy $\mathbb{E}[\|\mathbf{g}(\mathbf{x}_i)\|^2] < \infty$ where $\|\cdot\|$ denotes the Euclidean norm. Show that

$$\mathbb{E}_{\boldsymbol{\theta}_0}\left[\mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta})\right] = \mathbf{0} \text{ for all } \mathbf{g} \in \mathcal{G} \qquad \Longleftrightarrow \qquad \boldsymbol{\theta} = \boldsymbol{\theta}_0.$$

Consider the family of conditional Z-estimators $\widehat{\boldsymbol{\theta}}_n(\mathbf{g})$ for $\mathbf{g} \in \mathcal{G}$ satisfying

$$\frac{1}{n} \sum_{i=1}^{n} \mathbf{g}(\mathbf{x}_i) m\big(\mathbf{z}_i, \widehat{\boldsymbol{\theta}}_n(\mathbf{g})\big) = \mathbf{0}.$$

2. Suppose that $\boldsymbol{\Theta}$ is compact and $m(\mathbf{z}_i, \boldsymbol{\theta})$ is continuous in $\boldsymbol{\theta}$, uniformly in $\mathbf{z}_i$. Assume also that $\mathbf{g}$ satisfies $\boldsymbol{\theta} \neq \boldsymbol{\theta}_0 \implies \mathbb{E}_{\boldsymbol{\theta}_0}\left[\mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta})\right] \neq \mathbf{0}$. Show that $\widehat{\boldsymbol{\theta}}_n(\mathbf{g}) \to_{\mathbb{P}} \boldsymbol{\theta}_0$.

3. Suppose the conditions of part 2 hold and let $\boldsymbol{\theta}_0$ be in the interior of $\boldsymbol{\Theta}$. Assume further that $m(\mathbf{z}_i, \boldsymbol{\theta})$ is continuously differentiable in $\boldsymbol{\theta}$ with $\sup_{\mathbf{z}_i \in \mathbb{R}^{d_z}} \left\| \frac{\partial}{\partial \boldsymbol{\theta}} m(\mathbf{z}_i, \boldsymbol{\theta}) \right\| < \infty$ for all $\boldsymbol{\theta} \in \boldsymbol{\Theta}$. We impose these conditions for the remaining parts of this question. Show that there is a non-random influence function $\boldsymbol{\psi} : \mathbb{R}^{d_z} \times \mathbb{R}^{d_x} \times \mathbb{R}^d \times \mathcal{G} \to \mathbb{R}^d$ satisfying

$$\sqrt{n}\big(\widehat{\boldsymbol{\theta}}_n(\mathbf{g}) - \boldsymbol{\theta}_0\big) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \boldsymbol{\psi}(\mathbf{z}_i, \mathbf{x}_i; \boldsymbol{\theta}_0, \mathbf{g}) + o_{\mathbb{P}}(1).$$

   You may assume that any relevant matrices are invertible.

4. Show that

$$\sqrt{n}\big(\widehat{\boldsymbol{\theta}}_n(\mathbf{g}) - \boldsymbol{\theta}_0\big) \rightsquigarrow \mathcal{N}\big(0, \mathbf{V}(\boldsymbol{\theta}_0, \mathbf{g})\big),$$

   where $\mathbf{V}(\boldsymbol{\theta}, \mathbf{g}) = \boldsymbol{\Omega}(\boldsymbol{\theta}, \mathbf{g})^{-1} \boldsymbol{\Sigma}(\boldsymbol{\theta}, \mathbf{g}) \boldsymbol{\Omega}(\boldsymbol{\theta}, \mathbf{g})^{-\mathsf{T}}$ with

$$\boldsymbol{\Sigma}(\boldsymbol{\theta}, \mathbf{g}) = \mathbb{V}_{\boldsymbol{\theta}_0}[\mathbf{g}(\mathbf{x}_i) m(\mathbf{z}_i, \boldsymbol{\theta})] \qquad \text{and} \qquad \boldsymbol{\Omega}(\boldsymbol{\theta}, \mathbf{g}) = \mathbb{E}_{\boldsymbol{\theta}_0}\left[\mathbf{g}(\mathbf{x}_i) \frac{\partial}{\partial \boldsymbol{\theta}^{\mathsf{T}}} m(\mathbf{z}_i, \boldsymbol{\theta})\right],$$

   using the notation $\mathbf{A}^{-\mathsf{T}} = (\mathbf{A}^{-1})^{\mathsf{T}}$.

5. Propose an estimator of $\mathbf{V}(\boldsymbol{\theta}_0, \mathbf{g})$, denoted by $\widehat{\mathbf{V}}_n(\mathbf{g})$, and prove it is consistent.

6. Let $\mathbf{h} : \mathbb{R}^d \to \mathbb{R}^{d_h}$ be differentiable with $\frac{\partial \mathbf{h}(\boldsymbol{\theta}_0)}{\partial \theta}$ non-singular and assume $\mathbf{V}(\boldsymbol{\theta}_0, \mathbf{g})$ is invertible. Construct an asymptotically valid Wald-based hypothesis test for $\mathsf{H}_0 : \mathbf{h}(\boldsymbol{\theta}_0) = \mathbf{0}$ against $\mathsf{H}_1 : \mathbf{h}(\boldsymbol{\theta}_0) \neq \mathbf{0}$ at level $\alpha \in (0, 1)$.

7. Let $\mathbf{c} \in \mathbb{R}^d$ be a constant vector. Construct an asymptotically valid, heteroskedasticity-robust two-sided confidence interval for $\beta_0 = \mathbf{c}^\mathsf{T}\boldsymbol{\theta}_0$.

8. Show that for all $\mathbf{g} \in \mathcal{G}$,

$$\mathbf{V}(\boldsymbol{\theta}_0, \mathbf{g}) - \mathbf{V}(\boldsymbol{\theta}_0, \mathbf{g}^*) = \boldsymbol{\Omega}(\boldsymbol{\theta}_0, \mathbf{g})^{-1}\mathbb{E}_{\boldsymbol{\theta}_0}\big[\mathbf{U}\mathbf{U}^\mathsf{T}\big]\boldsymbol{\Omega}(\boldsymbol{\theta}_0, \mathbf{g})^{-\mathsf{T}},$$

   where

$$\mathbf{g}^*(\mathbf{x}_i; \boldsymbol{\theta}_0) = \frac{\mathbf{D}(\mathbf{x}_i; \boldsymbol{\theta}_0)}{\sigma^2(\mathbf{x}_i; \boldsymbol{\theta}_0)}, \quad \sigma^2(\mathbf{x}_i; \boldsymbol{\theta}) = \mathbb{V}_{\boldsymbol{\theta}_0}[m(\mathbf{z}_i, \boldsymbol{\theta}) \mid \mathbf{x}_i], \quad \mathbf{D}(\mathbf{x}_i; \boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{\theta}_0}\left[\frac{\partial}{\partial \boldsymbol{\theta}} m(\mathbf{z}_i, \boldsymbol{\theta}) \;\middle|\; \mathbf{x}_i\right],$$

   and

$$\mathbf{U} = \boldsymbol{\nu}_i - \boldsymbol{\Omega}(\boldsymbol{\theta}_0, \mathbf{g})\mathbb{E}_{\boldsymbol{\theta}_0}\big[\boldsymbol{\nu}_i^*\boldsymbol{\nu}_i^{*\mathsf{T}}\big]^{-1}\boldsymbol{\nu}_i^*, \quad \boldsymbol{\nu}_i = \mathbf{g}(\mathbf{x}_i)m(\mathbf{z}_i, \boldsymbol{\theta}_0), \quad \boldsymbol{\nu}_i^* = \mathbf{g}^*(\mathbf{x}_i; \boldsymbol{\theta}_0)m(\mathbf{z}_i, \boldsymbol{\theta}_0).$$

9. Show that under the semi-definite partial ordering defined by $\mathbf{V} \preceq \mathbf{V}'$ if and only if $\mathbf{V}' - \mathbf{V}$ is positive semi-definite, we have

$$\mathbf{g}^*(\mathbf{x}_i; \boldsymbol{\theta}_0) \in \operatorname*{arg\,min}_{g \in \mathcal{G}} \mathbf{V}(\boldsymbol{\theta}_0, \mathbf{g}).$$

   Propose a feasible, asymptotically efficient estimator of $\boldsymbol{\theta}_0$.

10. Suppose that $\mathbf{z}_i = (y_i, \mathbf{x}_i)$ takes values in $\mathbb{R} \times \mathbb{R}^{d_x}$ and assume that it satisfies $y_i = \mathbf{x}_i^\mathsf{T}\boldsymbol{\theta}_0 + \varepsilon_i$ with $\mathbb{E}_{\boldsymbol{\theta}_0}[\varepsilon_i \mid \mathbf{x}_i] = 0$ and $v(\mathbf{x}_i) = \mathbb{E}[\varepsilon_i^2 \mid \mathbf{x}_i]$. Consider the least-squares estimator $\widehat{\boldsymbol{\theta}}_{\mathrm{LS}}$ and the weighted least-squares estimator $\widehat{\boldsymbol{\theta}}_{\mathrm{WLS}}$ defined by the first order conditions

$$\frac{1}{n}\sum_{i=1}^n \mathbf{x}_i(y_i - \mathbf{x}_i^\mathsf{T}\widehat{\boldsymbol{\theta}}_{\mathrm{LS}}) = \mathbf{0} \qquad \text{and} \qquad \frac{1}{n}\sum_{i=1}^n \frac{\mathbf{x}_i}{v(\mathbf{x}_i)}(y_i - \mathbf{x}_i^\mathsf{T}\widehat{\boldsymbol{\theta}}_{\mathrm{WLS}}) = \mathbf{0}$$

    respectively. Compute the asymptotic variances of $\widehat{\boldsymbol{\theta}}_{\mathrm{LS}}$ and $\widehat{\boldsymbol{\theta}}_{\mathrm{WLS}}$ and discuss their relative asymptotic efficiencies. *Hint: You may wish to identify the optimal Z-estimator* $\widehat{\boldsymbol{\theta}}_n(\mathbf{g}^*)$.